GABRIEL GUARD

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EDUCATION

The University of Texas at Austin ***May 2030***
Bachelor of Science in Mechanical Engineering | Intended Minor in Robotics

EXPERIENCE

Longhorn Racing Electric | *Dynamics Mechanical Design Engineer* **Aug 2023 - May 2026**
- Executed ANSYS-validated topology optimization of brake pedal, reducing mass 52% (0.62 to 0.30 lb) under a 450 lbf pedal-face load; verified <1 mm deflection under off-axis loading and maintained structural safety margins
- Applied first-principles statics and iterative FEA to optimize pedal geometry for structural performance and manufacturability; designed for in-house 3-axis CNC milling
- Designed carbon fiber accelerator pedal faces from scratch, developing component geometry, mold designs, and laminate layups for composite fabrication

Throne Science | *R&D Engineering Intern* **Aug 2023 - May 2026**
- Designing a 4-DOF test rig integrating dual gantries, pivot mechanisms, fluid routing, motion control, and electronics to enable repeatable, position-controlled nozzle testing for AI audio prediction
- Owning mechanical and electrical system development, including component sourcing, custom CAD, fluid delivery, motion systems, and electronic enclosures to improve test fidelity and usability over a prior iteration
- Developing the rig for manufacturability, assembly, and shipment to an external testing facility, with emphasis on packaging constraints and repeatable test positioning

Competitive Robotics (FIRST FRC), Team 6800 | 1,020 hours **Aug 2023 - May 2026**
Mechanical Design Lead
- Designed four complete robot subsystems using top-down parametric CAD in Onshape; produced tolerance-driven engineering drawings and optimized assemblies for manufacturability, integration, and performance.
- Led end-to-end manufacturing and assembly of a robot subsystem, coordinating fabrication and inspection of 80+ parts across CNC milling, turning, routing, and powder coating; managed sourcing, inventory, BOMs, and dimensional verification against design tolerances while managing development in Jira
- Programmed CNC toolpaths in VCarve and Fusion 360, selecting tooling, feeds, speeds, and machining strategies to fabricate custom components using CNC routers, mills, manual lathes, and 3D printers
- Selected as primary competition driver, leading match strategy, coordinating alliance partners, and communicating under pressure to adapt strategy and execute coordinated match plans on the world stage

Umbra Design LLC | Lead Mechanical Design Engineer **Aug 2023 - May 2026**
- Founded an LLC to develop and commercialize proprietary translucent automotive sun visors, leading product design, prototyping, manufacturing, and assembly using 3D printing and laser cutting

Achievements
(Team) FIRST World Championship Division Winner, FIRST World Championship Division Finalist, FIRST In Texas District Championship Winner, FIRST In Texas District Championship Finalist, 4x District Event Winner, 6x Industrial Design Award Winner, 2x Excellence in Engineering, 1x Engineering Inspiration, (Individual) Team's Choice Award

PROJECTS

Valor CAD Challenge | 30 hours
- Organized an FTC design challenge with 117 participants, designing and modeling a mock FTC field, generating a game manual, and judging 28 submissions for practicality and feasibility.
- Produced and livestreamed a sponsored awards ceremony, reaching 750 viewers while coordinating award presentations and event logistics.

SKILLS

CAD & Analysis: SolidWorks, Onshape, Autodesk Inventor, Fusion 360, ANSYS
Certifications: Onshape Associate, Autodesk Inventor, Microsoft Office Associate (Excel, Word, PowerPoint), OSHA 10
Manufacturing CNC Mill, CNC Router, Lathe, Waterjet, Laser Cutting, FDM 3D Printing
Softwares & Skills: Jira, Confluence, Figma, Spanish (conversational)
Pursuing: SolidWorks Professional, Onshape Professional, MATLAB